Near-beam tagging of the far-forward lithium recoils at ePIC: what the aperture is worth, what the optics is worth, and whether superconducting nanowires help

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2026-08-26 · rewritten 2026-08-28 on the per-configuration Yellow Report divergences and the lithium tagging optics · figures from `evgen/scripts/nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py`, `nearbeam_sensor_budget.py` and `nearbeam_zid_power.py`; the far-forward geometry from `tools/fullsim` and the eic/epic main branch

Abstract. The two far-forward lithium tags of the programme — the intact ^{6}Li recoil of the coherent channel and the α spectator of the tagged channels — are rigidity-degenerate with the beam and are separated from it only by their angle at the interaction point, so the tagged fraction is an exponential in the square of the closest approach. This report prices that approach for each beam configuration against the three half-widths that compete to set it: the 10σ envelope of the Yellow Report optics (2.2, 1.8 and 0.92 mrad for ^{6}Li at 5 × 41, 10 × 100 and 18 × 275), the aperture of the ePIC Roman-Pot silicon measured by transporting an intact ^{6}Li through the geometry (2.0, 1.35 and 1.03 mrad in the September-2024 layout), and the envelope of a lithium tagging optics in which the horizontal β* is raised 50–176× with the pots following (0.33, 0.17 and 0.12 mrad). At the published optics the machine, not the sensor, binds at every configuration in the current geometry, and no closer approach by any technology recovers a tag: 7×10^{-8} to 6×10^{-27} of coherent recoils, and of the α spectators only the ≈ 1.5% off-rigidity slice that the Roman-Pot window catches at any optics. Under the tagging optics the silicon becomes the limit and a layer that reaches the envelope is what makes the optics worth having: 32–42% of coherent recoils at 1/7–1/13 of the luminosity, four clean |t| bins per configuration through the full reconstruction chain, and a 27–35% α tag. The gain lives in a strip 50–180 μrad wide at the inner edge. Superconducting nanowires do not improve either the approach or the charge identification: their dead edge is worth 0.005 mrad against a gap set by module tiling and by a machine-protection rule, and in charge identification the one bit they deliver is nearly sufficient in principle but the 50% geometric fill of the existing devices cannot reach 95% ^{6}Li efficiency over four planes, while the α + d breakup that the charge question exists to reject is two hits 5–150 mm apart in sensors ePIC already has.

1 · The angle is the measurement

The coherent measurement needs the intact ^{6}Li, whose A/Z is exactly the beam's; the tagged measurements need the α spectator, at 0.99813 of the beam rigidity. Neither is swept out by the dipoles, and the only handle is the transverse angle at the interaction point: the coherent recoil follows $\exp(-B|t|)$ with $B \approx 50\ \mathrm{GeV}^{-2}$ and $|t| = (\theta p)^2$, so the tagged fraction falls as the exponential of the square of the approach angle times the square of the beam momentum. The α's angular distribution is set by the cluster wave function, with a tail extending to a few hundred μrad at the top energy and a few mrad at the bottom (Figure 1b).

Three half-widths compete to set the approach, and until 2026-08-28 this programme confused two of them by applying one proton-derived, isotropic, energy-independent divergence of 72.7 μrad at every configuration. The **10σ envelope** is the operational retraction of the Roman Pots, a rectangle of half-widths $10(\sigma_h, \sigma_v)$ in angle: for ^{6}Li at the Yellow Report high-acceptance optics, 2.2 × 3.8, 1.8 × 1.8 and 0.92 × 0.92 mrad at 5 × 41, 10 × 100 and 18 × 275 (Report 3 §4.1) — a machine-protection and halo rule, not a sensor one. The **silicon aperture** is what the sensor package, its shield, its ASIC and its module tiling leave open, measured by transporting an intact ^{6}Li through the ePIC far-forward geometry (`tools/fullsim`): $|\theta_x| \gtrsim 2.00 / 1.35 / 1.03$ mrad against $|\theta_y| \gtrsim 3.0 / 3.0 / 2.3$ mrad, in a September-2024 layout that has since changed (§6). And the **tagging-optics envelope** is the 10σ rectangle of the lithium tagging optics of Report 1 §6.1, in which the horizontal β* alone is raised to the optimum of tagged fraction × luminosity: 0.33 × 3.8, 0.17 × 1.8 and 0.12 × 0.92 mrad at $r_h = \beta^*_x/\beta^*_{x,HA} = 49.7$, 175.6 and 89.3 and $L/L_{HA} = 1/7.1$, 1/13.3 and 1/9.5

(`farforward.tagging_optics_point`). In the aperture scan of §2 the vertical half-width is the larger of the measured aperture and the envelope, per configuration; in the chain of §2.1 the pots at the silicon use the measured rectangle and the pots that follow use the tagging optics' 10σ rectangle in both planes, as Report 1 §6.1 prices it.

2 · What a closer approach is worth

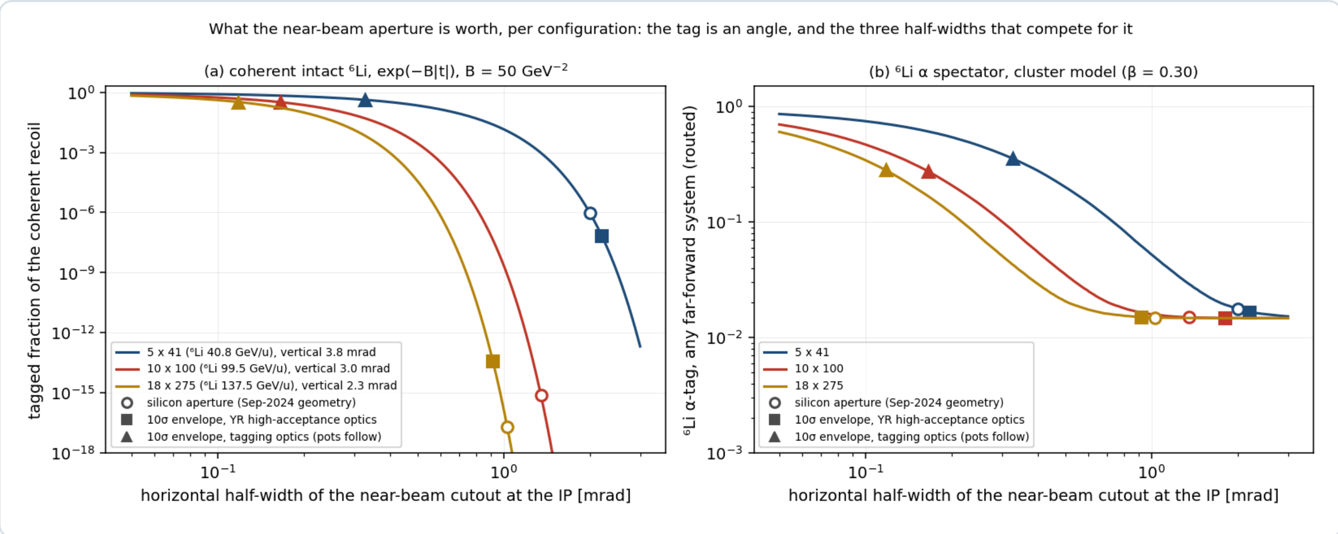

Figure 1 — the price of every aperture. Tagged fraction of the coherent intact ⁶Li (a, exp(−B|t|), B = 50 GeV⁻²) and of the ⁶Li α spectator routed through the far-forward systems (b, cluster model, β = 0.30) against the horizontal half-width of the near-beam cutout in angle, per configuration, with the vertical half-width fixed as in §1. Open circles: the silicon aperture of the September-2024 geometry; squares: the 10σ envelope of the Yellow Report high-acceptance optics; triangles: the envelope of the tagging optics. The curves are technology-independent and price any closer approach by any means; the markers are where the three constraints sit. `nearbeam_aperture_scan.py` .

CONFIGURATION (⁶LI GEV/U)	HALF-WIDTH	5 × 41 (40.8)	10 × 100 (99.5)	18 × 275 (137.5)
coherent recoil, tagged fraction	silicon, Sep-2024 (2.00 / 1.35 / 1.03 mrad)	9.8×10^{-7}	7.7×10^{-16}	1.9×10^{-17}
	YR high-acceptance envelope (2.20 / 1.80 / 0.92)	7.2×10^{-8}	6.2×10^{-27}	3.9×10^{-14}
	tagging-optics envelope (0.33 / 0.17 / 0.12)	**0.42**	**0.32**	**0.33**
⁶Li α spectator, tagged (any far-forward system)	silicon	0.018	0.015	0.015
	YR high-acceptance envelope	0.017	0.015	0.015
	tagging-optics envelope	**0.35**	**0.27**	**0.28**

Table 1 — the three half-widths priced (Figure 1's markers; `nearbeam_aperture_scan.py` , 2026-08-28). The α row is the tag routed through the far-forward systems (`farforward.acceptance_summary`), the same quantity as Report 3 Table 6: the ≈ 1.5% off-rigidity slice below R = 0.95 is tagged at any half-width, and the rest is the R ≈ 1 tail outside the rectangle. The tagging optics is paid for in luminosity, 1/7.1, 1/13.3 and 1/9.5 of the high-acceptance value, so its rows are to be read as yield per unit high-acceptance luminosity ×0.14, ×0.075 and ×0.106.

Two statements follow from the table and neither depends on a sensor. At the Yellow Report optics the envelope is at or inside the silicon at 5 × 41 and 10 × 100, and inside it at 18 × 275 in the current geometry (§6): the machine binds, and moving the silicon closer buys nothing at any configuration — the coherent tag stays below 10^{-7} and the α tag at the 1.5% of its off-rigidity slice. At the tagging optics the envelope shrinks by 7–11× and the silicon, still at 1.0–2.0 mrad, becomes the only limit: pots that follow the envelope tag 32–42% of coherent recoils and 27–35% of α spectators, pots that stay where they are tag the off-rigidity slice alone. The layer and the optics are multiplicative levers, and the rest of this report prices the layer given the optics.

The top energy is no longer excluded. Under the single 0.727 mrad envelope of the earlier analysis the coherent channel at 18×275 required $p_T > 0.60$ GeV and was dead at 10^{-9} for any aperture; at the tagging optics the envelope is 0.12 mrad, $p_T > 0.10$ GeV, and the top configuration is the most productive of the three at equal luminosity. The earlier ordering of the configurations was an artefact of the divergence used.

2.1 · Through the chain

Acceptance is not the measurement. Figure 2 runs the full coherent chain of Report 2 §4.5 — the Roman-Pot emulation with the anisotropic divergence and the rectangular cutout, the two azimuths $\alpha = \varphi_e - \varphi_S$ and $\beta = \varphi_t - \varphi_S$, the spin-state-sorted two-dimensional harmonic fit — at the tagging optics of each configuration, once with the pots fixed at the measured silicon aperture and once with the pots following the envelope, at the tagging optics' luminosity (10 fb^{-1}/u per year × L/L_{HA}) and $P_{zz} = 0.60$.

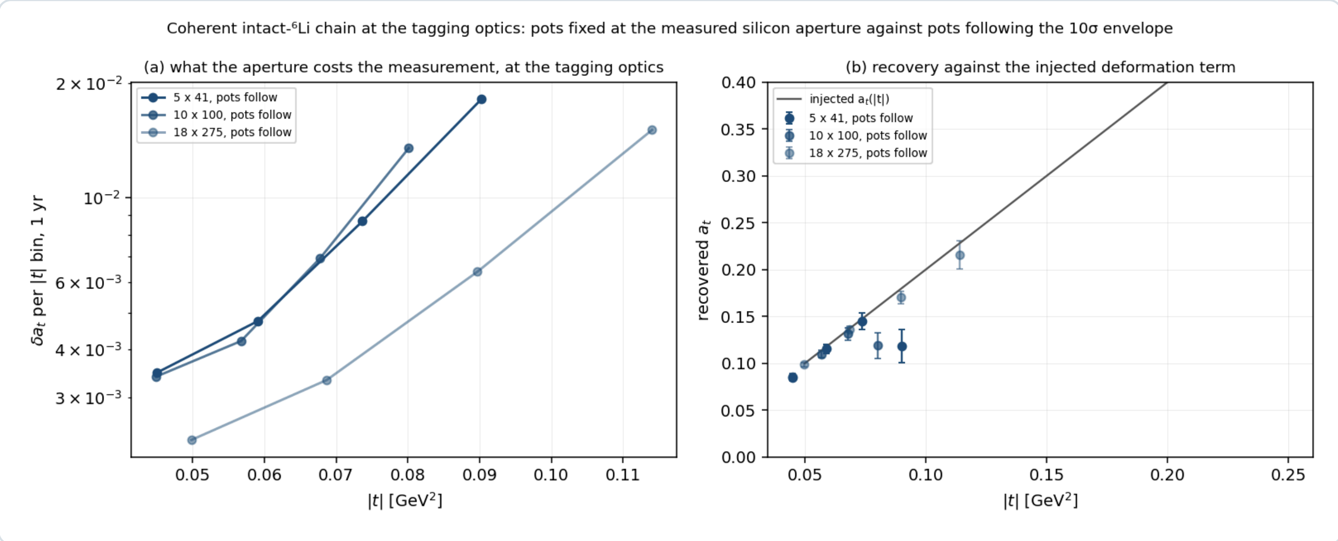

Figure 2 — the chain at the tagging optics. (a) δa_t per $|t|$ bin at one year with the pots following the envelope, for the three configurations; the silicon-aperture runs return no bins and do not appear. (b) The recovered a_t against the injected deformation term. `nearbeam_reach_gain.py`, 2×10^6 Monte-Carlo recoils per run.

Configuration	Pots	Cutout [mrad]	Acceptance	N_{tag} / yr	$\|t\|$ bins	δa_t per bin	a_e (injected 0.0100)
5 × 41	silicon	2.00 × 3.80	0	0	0 of 4	—	—
5 × 41	follow	0.33 × 3.80	0.41	2.5×10^6	4 of 4	0.0035, 0.0047, 0.0087, 0.018	0.0102 ± 0.0026 … 0.0136 ± 0.0140
10 × 100	silicon	1.35 × 3.00	0	0	0 of 4	—	—
10 × 100	follow	0.17 × 1.80	0.31	2.9×10^6	4 of 4	0.0034, 0.0042, 0.0070, 0.0135	0.0102 ± 0.0022 … 0.0159 ± 0.0097
18 × 275	silicon	1.03 × 2.30	0	0	0 of 4	—	—
18 × 275	follow	0.12 × 0.92	0.32	6.0×10^6	4 of 4	0.0023, 0.0033, 0.0064, 0.0151	0.0102 ± 0.0016 … 0.0004 ± 0.0117

Table 2 — what the aperture costs the measurement, at the tagging optics. The four $|t|$ bins are 0.05–0.08, 0.08–0.12, 0.12–0.17 and 0.17–0.25 GeV²; a_e is the exotic-glue coefficient, recovered consistent with its injected value in every bin. The recovered a_t falls below the injected curve in the highest bin at 5 × 41 and 10 × 100 (0.118 ± 0.018 against 0.181; 0.119 ± 0.014 against 0.160), a bin-centring effect of the steep spectrum at the cutout's edge rather than a bias of the fit; the three lower bins agree to within 1.5σ, the largest pulls (1.5σ, 1.3σ, 1.4σ) all on the low side. The analytic yields of Report 1 §6.1, 2.6, 3.0 and 6.1 × 10⁶ per year, are reproduced to 5% by the Monte-Carlo rectangle.

With the pots fixed the measurement does not exist at any configuration: the silicon is 5–8 envelope widths out and the acceptance is zero to the precision of 2×10^6 recoils. With the pots following, every configuration returns four |t| bins with δa_t = 0.002–0.005 in the two lowest, which is the sensitivity Report 1 §6.1 prices at 5σ floors of 1.7, 2.1 and 1.6% per unit P_{zz} per year and 2.8, 4.4 and 2.6 years to 5σ on a 1% exotic-glue term. The conclusion of Report 2 that the coherent channel is a low-energy programme was conditioned on the single 0.727 mrad envelope; at the tagging optics the top configuration is the best covered.

> **A trap the earlier analysis exposed and this one removes.** The reconstructed-level coherent money plot carried `cut_scale_x = 2.5` from a pre-measurement belief that the pots surround a wide horizontal slot; with the aperture measured, keeping it imposed a 25σ retraction no machine rule asks for, binding before the geometry did. Every script of this report holds the envelope at 10σ per axis, applies the rectangle to each fragment's azimuth, and lets the measured geometry be the only other aperture.

3 · Where the gain lives, and why the dead edge is not it

The exponential piles the gain against the inner edge. Moving the horizontal half-width from the measured silicon to the tagging envelope, half of the recovered coherent yield lies within 183, 69 and 50 μrad of the envelope at the three configurations, 90% within 504, 194 and 141 μrad, and 99% within 842, 328 and 239 μrad (`nearbeam_sensor_budget.py`); for the α spectator at 137.5 GeV/u the same widths are 77, 269 and 624 μrad. A near-beam insert is a narrow strip, not a plane: at R_{12} = 30.6 m the 50% strip at 18 × 275 is 1.5 mm wide, and a 3 mm strip along 20 mm of vertical extent on both sides of the beam is 120 mm² — 960 channels of 1 × 1 mm² microwire tiles, or 133 000 nanowire devices of 30 × 30 μm².

Near-beam recoil tagging is on record as the first of the four EIC applications of superconducting nanowires listed in Yellow Report §14.5.3 — "a Roman pot detector in the forward region about 35 meters or more from the interaction point to tag low momentum transfer recoiling ions" [1] — and the FY22 generic-detector R&D proposal states the mechanism: "The so-called 10σ rule-of-thumb for Roman pots prevents the beam from damaging the detectors by limiting its proximity to the beam. Radiation hard nanowire detectors will be able to move closer to the beam, thus extending the acceptance reach at lower t" [2]. Neither half of that mechanism survives the arithmetic. The 10σ rule protects the beam and keeps the sensor out of the halo — the ePIC geometry file itself calls its per-energy insertions "the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have" [3] — and a more radiation-hard sensor does not relax it; the LHC ran the CT-PPS pots at 25σ at 6.5 TeV. And the dead-edge argument, while it describes a real sensor property — a nanowire's active area reaches within a few hundred nanometres of the substrate edge [1], sharper than the 10 μm telescope used to look at it [8] — is worth ≈ 150 μm against the 100–200 μm of the slim-edge planar and 3D silicon already deployed in Roman Pots: 0.005 mrad at R_{12} = 30.6 m, 4% of the tagging envelope at the top energy and 0.5% of the gap between the silicon and that envelope. What does survive is the granularity half of the argument, and it belongs to ePIC rather than to a detector-R&D group: the insertion is quantised by the module, a 16 mm block against a tagging envelope of 3.6 mm at 18 × 275, and the fix is a staircase retraction, narrower modules, or the x-moving layer ePIC is already designing — "one 'x' moving layer and one 'y' moving layer in each station" [5]. That layer, following the envelope of the tagging optics, is the entire content of Table 2.

4 · Charge identification

An intact ^{6}Li, an α and a d from breakup share rigidity and velocity, so only dE/dx separates them, as z² (9 : 4 : 1); no EIC document addresses Z identification in the pots, and the only documented concept is a z² Cherenkov behind the IR-8 secondary-focus pots. Open question #19 of the programme asks whether the pots can tell an intact ^{6}Li from its breakup.

A nanowire will not do it by pulse height, because the device latches: the amplitude is the diverted bias current, and both Fermilab/JPL beam papers find the amplitude distributions of 120 GeV hadrons and muons, 8 GeV pions and showering electrons indistinguishable [7,8]. **It would do it by firing threshold.** In the normal-core hot-spot regime a deposit Q makes a normal core of radius $r_s \propto \sqrt{Q}$ and the wire switches when the core spans it against the bias-suppressed superconducting width,

$$r_s = \sqrt{\frac{Q}{e\,\pi\,c\,\rho\,(T_c - T_0)}}\,, \qquad \frac{I_{\rm th}}{I_c} = 1 - \frac{2r_s}{w}$$

a relation three groups reach from three directions — Argonne's Eqs. 1–2 for 120 GeV protons [4], Renema's photon form, and the ion literature's $I_{th}/I_c = 1 - (zeV)^{½}C/w$ [9]. Because $dE/dx \propto z^2$ at fixed velocity, and a ^{6}Li at 137.5 GeV/u has βγ = 148 against 128 for Argonne's calibration proton, r_s scales linearly in z with no velocity correction: 134 nm (p, d), 268 nm (α), 402 nm (^{6}Li) on Argonne's anchor.

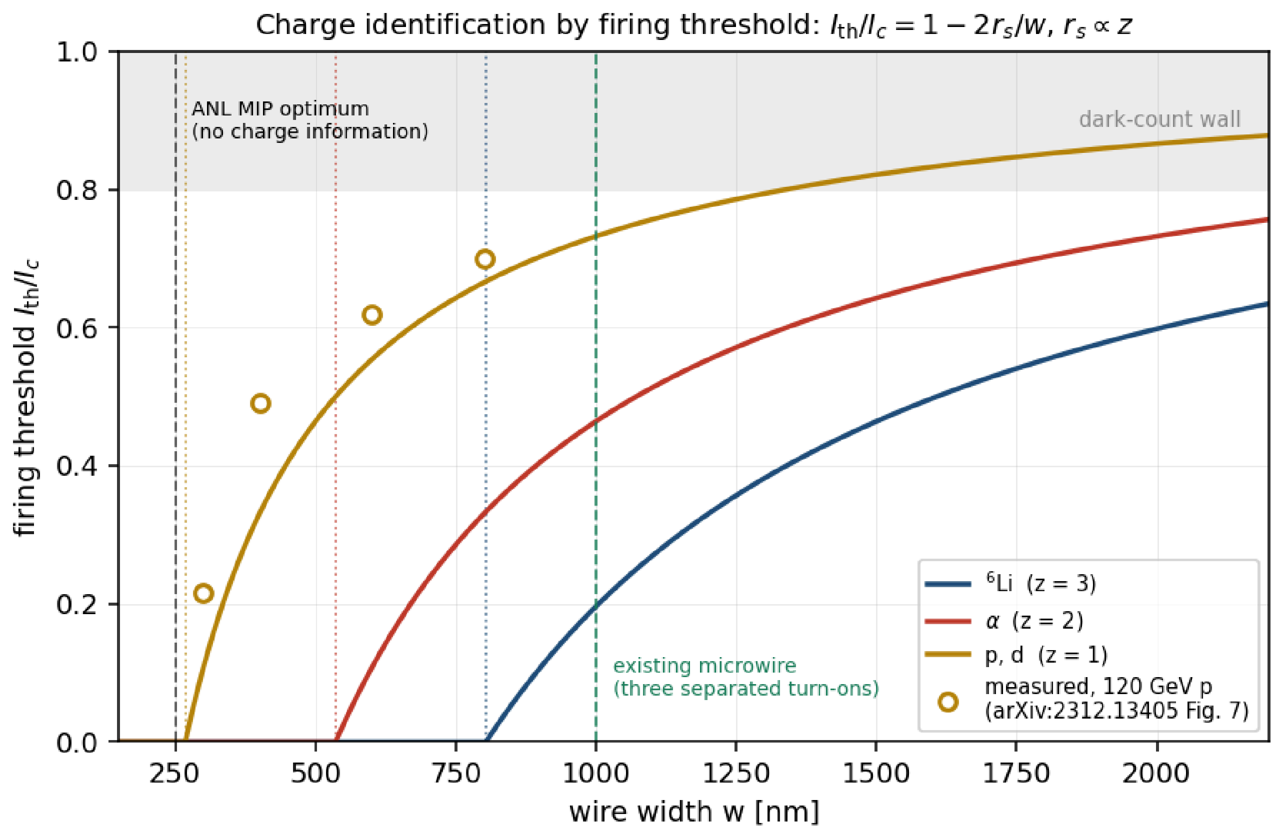

Figure 3 — Z by firing threshold. I_{th}/I_c against wire width for the three species from $r_s \propto z$ anchored on Argonne's 134 nm; open circles are their measured thresholds, digitised from Fig. 7 of [4]; dotted verticals mark $w = 2r_s$, below which the wire fires at any bias and carries no charge information. Inverting the measured points individually gives r_s = 102–118 nm (mean 113) against the quoted 134 nm; every design number is a ratio between species and the offset moves nothing (`evgen/tests/test_nearbeam.py`). `nearbeam_sensor_budget.py`.

WIRE WIDTH	I_{TH}/I_C, P/D	A	^{6}LI	SEPARATED TURN-ONS
250 nm	0.00	0.00	0.00	**none** — Argonne's MIP optimum
400 nm	0.33	0.00	0.00	**none**
800 nm	0.67	0.33	0.00	**α from d**
1000 nm	0.73	0.46	0.20	**all three** — the existing microwire
1500 nm	0.82	0.64	0.46	**all three**

Table 3 — turn-on bias per species and wire width. Argonne's efficiency optimum, w ≈ 250 nm, is $2r_s$ for a proton: maximal MIP efficiency and zero charge information. Z identification needs deliberately wide wires — the 1 μm microwire Caltech/JPL/Fermilab already fabricate [7] — where two bias points at 0.33 and 0.60 I_c tag Z by the firing pattern alone,

both below the 0.80 I_c dark-count wall and below the photon threshold, which is what a device looking at a 300 K beam pipe needs anyway.

Three caveats weigh on the model. r_s = 134 nm is the extrapolated zero-crossing of a four-point fit whose authors write that "the physical validity of this simple model is a question of future work" [4]; the volume in which Q is counted differs by 5000 between Argonne's two papers — the 20 eV film deposit in 2024, the 0.1 MeV substrate deposit in 2026 [5] — which leaves $r_s \propto z$ intact but not the absolute radii; and nobody has varied Z at fixed velocity on one of these devices, Argonne's ^{241}Am α differing from their proton through $1/\beta^2$ and every ion demonstration entering charge as $E = zeV$ rather than as z^2 in dE/dx.

4.1 · Against the incumbent: bits versus fill factor

EICROC digitises, per channel, an 8-bit charge from a 40 MHz successive-approximation ADC alongside its 25 ps time-of-arrival, behind an AC-LGAD with a 30 µm active thickness at 500 µm pitch [11]. The right figure of merit is not a separation in σ — a Landau upper tail falls as 1/λ, nothing like a Gaussian — but a fake rate at matched efficiency, with the Landau sampled (`nearbeam_zid_power.py`: four planes, 95% ^{6}Li efficiency, 1.5×10^6 events, 6.7×10^{-7} statistical floor):

READOUT	Α FAKE RATE	^{6}LI EFFICIENCY
8-bit charge, per-plane likelihood ratio — the Neyman–Pearson optimum	2.3×10^{-5}	0.950
one bit per plane, majority-of-k, 99% efficient planes	3.1×10^{-5}	0.950
truncated mean — the standard analogue dE/dx estimator	2.7×10^{-3}	0.950
plain sum of the four planes	5.3×10^{-2}	0.950

Table 4 — fake rate at matched efficiency. One bit costs a factor 1.4 against the optimum; a truncated mean costs a factor 86 against one bit, because a Landau has no mean and a single δ ray drags it. The species are far apart (MPV 31.7 against 75.2 keV) and the discriminating power comes from requiring coincidence across planes, not from precision within one.

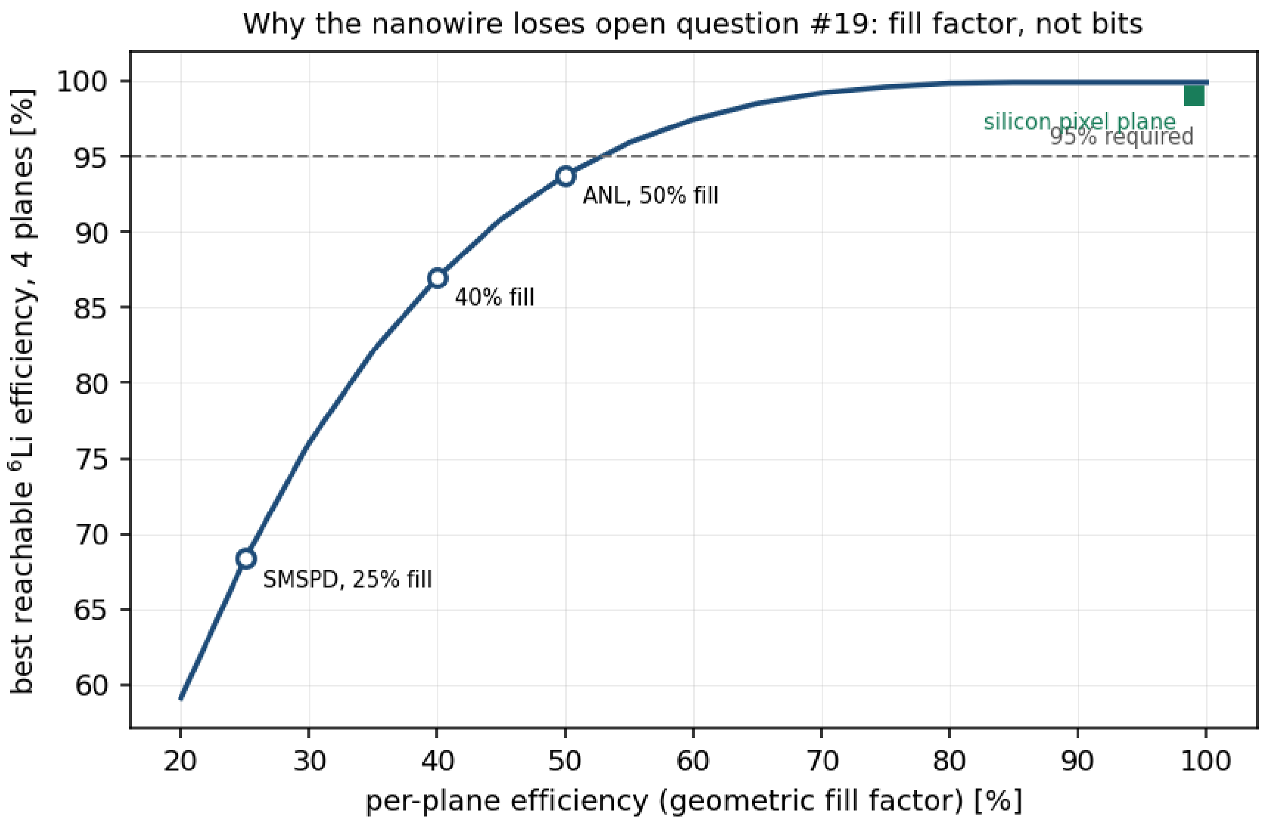

Figure 4 — fill factor, not bits. The coincidence that makes one bit sufficient needs every plane to record the track. A silicon pixel plane does so ≈ 99% of the time; a superconducting wire comb only over its geometric fill — 25% (SMSPD [7]), 40% [8], 50% (the Argonne EIC-targeted device [4]). At 50% a four-plane array tops out at 93.7% ^{6}Li efficiency and cannot reach 95% at any working point. `nearbeam_zid_power.py`.

The nanowire therefore loses on a fabrication number rather than an information-theoretic one, which is the one item on this list the Argonne group could change. The unknown that outranks the comparison is the incumbent's own: EICROC's 8-bit full-scale range is unpublished. A ^{6}Li deposits 75 keV = 3.3 fC unamplified, ≈ 67 fC at LGAD gain 20 (α: 28 fC), while EICROC0 is documented as "sensitive to low charge (2 fC)" and characterised at 12.5 fC; if the pot front end is MIP-ranged the ^{6}Li clips and α and ^{6}Li pile up at full scale, and LGAD gain suppression for highly ionising particles, measured at up to ≈ 2.5×, compresses them toward each other besides (Report 3 Table 9, item 5).

4.2 · The limit on any dE/dx scheme, and the handle that is free

The δ-ray upper tail is the same for every technology and nearly independent of sample thickness; thickening a plane does not help, independent planes do, because an α needs four upward fluctuations rather than one — which is why the four-plane rates of Table 4 sit at 10^{-5} while any single plane is at the percent level. The rates are conservative: the Landau is sampled untruncated, but a δ ray above ≈ 50 keV escapes a 30 μm layer.

The background #19 exists to reject, ^{6}Li → α + d, is separable without dE/dx at all. Both fragments sit at beam rigidity, but the breakup's relative momentum is transverse and unboosted, and the α at $4p_u$ and the d at $2p_u$ take opposite kicks of the same k_T. Sampled from the repository's own bound-state density (κ = 60.7 MeV/c; the 3^+ resonance's 42.2 MeV/c is a different production mechanism), the separation at the pots is

CONFIGURATION	MEDIAN SEPARATION	16–84%	IN 500 MM PIXELS	SINGLE-FRAGMENT LOSS PER BREAKUP	MERGE PROBABILITY
18 × 275	10.9 mm	4.9–22.7	22	0.9%	≤ 0.26%
10 × 100	30.1 mm	13.4–62.3	60	8.8%	≤ 0.035%

5 × 41	73.4 mm	32.6–152	147	27%	≤ 0.006%

Table 5 — the α + d topology at the pots, with R_{12} = 30.6 m measured at 18 × 275 and carried to the other rows for want of the per-optics value. An intact ^{6}Li is one hit; the breakup is two, tens of pixels apart, in sensors that exist. The commoner failure is not a merge but a partner leaving the sensor, at 5 × 41 past the pot's half-height; a merged pair deposits $\Sigma z^2 = 5$ against 9 and the charge readout separates it at ≈ 4σ per plane.

5 · What stands in the way of a cryogenic sensor in a pot

OBSTACLE	THE NUMBER	STATUS
cold-stage power	NbN needs 2.8–3 K, a WSi microwire 0.8 K; a 4 K pulse tube delivers 1.5 W at 4.2 K and nothing at its 2.8 K floor, on a 190 kg compressor drawing 10.7 kW	hard
beam-induced RF heating	a ferrite-shielded TOTEM pot dissipates ≈ 40 W at the LHC; scaling $Q^2M/\sigma_z^{3/2}$ to the EIC gives ≈ 16 / 8 / 0.8 W at highest-E_{CM} / highest-L / 41 GeV — even the last exceeds a pulse tube's whole 4.2 K lift	hard
in-vacuum precedent	none for a biased superconducting detector inside an accelerator's beam vacuum; the CryoBLM and AD SQUID precedents sit outside it, and the AD beam is ≈ 10^7 antiprotons against 8 × 10^{13} protons	open
vibration	cold-head displacements 7–9 µm, 10 nm with a decoupling structure that is heavy and stiff — the opposite of a pot on motorised rails	mitigable
radiation hardness	unmeasured: the FY22 deliverable was "an upper limit for the accumulated dose … currently unknown", and no publication reports the LEAF irradiation or the Hall C run; the 2023 Quantum Sensors for HEP report says "published studies in this area are lacking"	unmeasured
area	the largest device run in a GeV beam is 2 × 2 mm² with 8 channels [8]; the 120 mm² strip of §3 is 30× that, the whole gap 1000×	funded R&D
latching	a hard failure, not a dead time; Argonne's ^{241}Am run — a ≈ 1 µm hot spot in a 100–200 nm wire, clean plateaus — is the reassurance [5]	addressed
rate	a few Hz per channel in normal running, 30–50 kHz at ≈ 10σ from halo [6], four orders below the ≈ 10^8 s^{-1} a nanowire's fall time allows	closed

Table 6 — obstacles. None is a physics objection; all are load-bearing. Cryogenics also inverts the packaging argument: windowless operation in the beam vacuum and an edge a few hundred nanometres from the substrate rim are exactly what a near-beam sensor wants, but a device that cannot see a 300 K surface (≈ 3 × 10^{18} photons cm^{-2} s^{-1} above its 0.04 eV threshold) must sit behind a cold shield, which is material in front of the sensor; biasing below the photon threshold resolves it in principle and has never been demonstrated next to a beam.

6 · The ePIC pot geometry has moved

The aperture of §1 was measured in the September-2024 `epic-main`. The current main branch [3] tiles the pots with 16 × 16 mm modules in place of 32 × 32, carries per-energy 10σ offsets in `beamline_*.xml` in place of an energy-independent insertion, and has the 1 mm aluminium RF shields commented out ("we don't know if we will even need it", October 2025); the implied blind block at 18 × 275 is 16 mm (x) × 2.7 mm (y) against 32 × 7. The old block gives 32 mm / 30.6 m = 1.046 mrad against the 1.03 mrad the transport measured, the 1.5% agreement being the check that the measurement and the file reading confirm each other.

The correction runs in opposite directions at the two ends. At 18 × 275 the 16 mm block implies ≈ 0.52 mrad, inside the 0.92 mrad Yellow Report envelope — so at the published optics the beam binds there too, and the machine is the only constraint at all three configurations — but four times the 0.12 mrad tagging envelope, so under the tagging optics the silicon binds again until a layer follows. At 5 × 41 the per-energy insertion moves the inner edge out to 29.6 mm, because the pots now retract for the larger low-energy beam, and the silicon aperture there gets worse. Every aperture-conditional number of this programme is conditioned on the September-2024 measurement and should be re-measured against the current release; the curves of Figure 1 are unaffected, which is why they are plotted against the half-width, and only the marker labelled silicon moves. The per-optics R_{12} is measured for 18 × 275 alone, so this report works in angle and quotes millimetres only there.

7 · Recovering the tag: beam and detector

The levers available at IP6, priced at 5 × 41 against the Yellow Report baseline, are unequal. A pot standoff of 8σ rather than 10σ is worth ×230 at no luminosity, against a halo rate and an aperture hierarchy that do not yet exist and a RHIC precedent of 8σ for the pot with the active edge at 10σ. Detector relocation along the far-forward line is worth ≤ ×1.75. A lower beam energy is exhausted at γ = 43.5; a colder beam from lower intensity runs backwards, since fixed-rigidity injection puts an A/Z = 2 ion at half the proton's γ with ≈ 2.5× the space charge and there is no store cooling in the baseline to settle into; dispersive tagging at IP6 is 30–60× short of the 0.1 rigidity threshold of Gamage et al. [12] and the beam pipe cannot be instrumented past ≈ 30 m because of the crab cavities; and the LHC's "3σ" is a pot window with the silicon at 4.8–5.4σ in fills of 3×10^{11} protons, not an active edge. The lever that closes is β*, and Report 1 §6.1 prices it in the form this report has used throughout: the horizontal plane alone, the pots following, at r_h = 50, 176 and 89 and 1/7.1, 1/13.3 and 1/9.5 of the luminosity, yielding 2.5, 2.9 and 6.0 × 10^6 tagged recoils per year (Table 2), an 8–11σ measurement of the shape term per year and 5σ on a 1% exotic-glue term in 2.8, 4.4 and 2.6 years. A second interaction region's secondary focus at ≈ 20% efficiency would be worth 3×, 8× and 6× that yield at equal luminosity, a luminosity that is not published.

The energy ordering is now the reverse of what the earlier analysis found, and for the same reason: at the published optics the low-energy configuration led by five orders because a fixed 0.727 mrad envelope is the smallest fraction of the 41 GeV/u recoil's angular scale, while at the tagging optics each configuration's envelope shrinks with its own divergence and the top configuration, with four times the coherent events at equal luminosity, leads. Both orderings are carried by proton luminosities applied to a lithium beam, and with a species-limited luminosity the ranking is open. The two questions for C-AD are those of Report 3 Table 9: can IR6 be matched with the horizontal hadron β* raised 50–180× and the electron β* co-de-squeezed so ξ_p stays ≤ 0.015, with what IP-to-pot phase advances; and what physically sets the 10σ retraction. The published high-acceptance β* factors shrink toward low energy — 5.21 at 18 × 275, 1.49 at 10 × 100, 1.00 at 5 × 41, where the two optics tables are identical — so the headroom the designers found runs out where the smallest absolute ask, β^*_x ≈ 45 m, sits.

8 · Conclusion

The far-forward lithium tags are an angle, and the angle is set by the machine before it is set by the sensor. At every published optics the 10σ envelope is at or inside the silicon in the current geometry, no closer approach by any technology tags a coherent recoil or more than the 1.5% off-rigidity slice of the α spectators, and the earlier finding that a closer approach was worth ×26 to ×569 was an artefact of a proton divergence applied to a lithium beam. Under a lithium tagging optics the envelope shrinks by 7–11× and the silicon, unmoved, becomes the whole limit: a layer that follows the envelope tags 32–42% of coherent recoils and 27–35% of α spectators at 1/7–1/13 of the luminosity, gives four clean |t| bins per configuration through the full reconstruction chain, and its gain lives in a strip 50–180 μrad wide at the inner edge. The layer and the optics are multiplicative, and neither is worth anything alone.

Superconducting nanowires appear nowhere in that layer. The near-beam role fails because a sensor's dead edge is worth 0.005 mrad against a gap set by module tiling and by a rule that protects the machine, and the charge-identification role fails because the incumbent AC-LGAD already digitises an 8-bit charge across four planes, because the nanowire's one bit is nearly sufficient in principle but its 50% geometric fill cannot reach 95% efficiency, and because the α + d breakup is two hits tens of pixels apart. That is a conclusion about this application, not the technology: the field result that makes it unique — saturated efficiency to 5 T parallel to the device plane [10] — is worth nothing in a field-free drift and everything inside a magnet. What would help the lithium tags, in order: the tagging optics; the x-moving near-beam layer ePIC is already designing, following it; the charge EICROC already digitises, with its dynamic range published; two-hit topology for the α + d channel; and the IR-8 secondary focus.

References

[1] R. Abdul Khalek et al., *Science Requirements and Detector Concepts for the Electron-Ion Collider: EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419, §14.5.3 (`refs/2103.05419_part4.pdf`): the four EIC SNSPD applications; edgeless operation "within a few 100 nm of the substrate edge"; and Tables 10.1–10.2, the beam divergences per configuration and optics.

[2] W. Armstrong (PI) et al., *Superconducting Nanowire Detectors for the EIC*, FY22 EIC-Related Generic Detector R&D proposal #18, §3.3 Concept 1, reviewed and partially funded (Report of the 2022 EIC-related Generic Detector R&D Review Committee, June 2023).

[3] eic/epic, branch main: `compact/far_forward/roman_pots_eRD24_design.xml` and `compact/fields/beamline_{18x275,10x100,5x41}.xml`, read directly 2026-08-26. Module 1.6 × 1.6 cm; per-energy offsets with the comment "These are the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have. They are not perfectly ten-sigma for reasons of physical geometry."; RF-shield components commented out, "Oct. 2025".

[4] S. Lee, T. Polakovic, W. Armstrong, A. Dibos, T. Draher, N. Pastika, Z.-E. Meziani, V. Novosad, *Beam Tests of SNSPDs with 120 GeV Protons*, Nucl. Instrum. Meth. A 1069 (2024) 169956, arXiv:2312.13405 (`refs/2312.13405.pdf`): 12 nm NbN, 30 × 30 μm², wire widths 300–800 nm at 2.82 K; Eqs. 1–2; r_s = 134 nm; the ≈ 250 nm optimum; background rising exponentially above I_b/I_c = 0.8.

[5] S. Lee, W. Armstrong, J. DiPreta, C. Dulya, V. Novosad, T. Polakovic, *Optimization of Cryogenic Detector Test Station by Rejecting Electromagnetic Interference*, Nucl. Instrum. Meth. A 1093 (2027) 171953, arXiv:2601.03158 (`refs/2601.03158.pdf`): the 5.5 MeV ^{241}Am α at a ≈ 1 μm hot spot; clean plateaus with no latching. And A. Jentsch, ePIC Collaboration Meeting, JLab, July 2025: the x- and y-moving layers.

[6] ePIC Collaboration, *Preliminary Design Report*, 29 September 2024, the Roman Pots and off-momentum detectors section: "Rates during normal operations, with expected vacuum of 10^{-9} mbar, are a few Hz/channel. However, the beam halo could potentially provide rates of 30–50 kHz at ~10σ from experience of Roman pots at STAR."

[7] C. Wang et al. (Caltech / JPL / Fermilab), *Towards High-Efficiency Particle Detection Using Superconducting Microwire Arrays*, arXiv:2510.11725 (`refs/2510.11725.pdf`): 8-channel 1 × 1 mm² WSi, 1 μm wires on a 3 μm gap (25% fill), 0.8 K; 75% fill-normalised efficiency and 130 ± 17 ps at CERN SPS H6.

[8] C. Peña, C. Wang, S. Xie et al., *Characterization of a Superconducting Microwire Single-Photon Detector Array for Charged-Particle Detection*, JINST 20 (2025) P03001, arXiv:2410.00251 (`refs/2410.00251.pdf`): 2 × 2 mm², 8 channels, 30 ± 1 μm spatial resolution; amplitude distributions indistinguishable across species; edge sharper than the 10 μm telescope.

[9] R. Cristiano, M. Ejrnaes, A. Casaburi, N. Zen, M. Ohkubo, *Superconducting nano-strip particle detectors*, Supercond. Sci. Technol. 28 (2015) 124004; Suzuki et al., Rapid Commun. Mass Spectrom. 24 (2010) 3290; US Patent 8,872,109 B2; J. J. Renema et al., arXiv:1301.3337.

[10] T. Polakovic et al., *Superconducting nanowires as high-rate photon detectors in strong magnetic fields*, Nucl. Instrum. Meth. A 959 (2020) 163543, arXiv:1907.13059 (`refs/1907.13059.pdf`): saturated efficiency to 5 T parallel to the device plane, ≈ 0.5 T perpendicular; 10^7 counts/s measured.

[11] EICROC0 characterisation and the ePIC AC-LGAD specification: A. Jentsch, DIS 2023 (`refs/ePIC_far_forward_talk_DIS_2023_v2.pdf`); OMEGA/IJCLab EICROC documentation as cited in Report 3 §6.1.

[12] B. R. Gamage et al., *Simulation of light ion fragment detection at the EIC*, arXiv:2105.13564, Table 1: the minimum detectable rigidity offset.

[13] This repository: Report 1 §6.1 and `evgen/scripts/tagging_optics.py` (the tagging optics priced); Report 2 §4.5 (the coherent chain); Report 3 §4–5 (the divergence derived, the tags per optics); `plans/09_nearbeam_nanowire_far_forward.md` and `plans/10_beam_divergence_light_ions.md`.

Appendix A · Revision history

DATE	REVISION
2026-08-28	Rewritten as a paper on the per-configuration Yellow Report divergences of Report 3 §4.1 and the lithium tagging optics of Report 1 §6.1, replacing the single proton-derived 72.7 μrad envelope that every number before 2026-08-27 used. The verdict on the near-beam layer changes: at the published optics the machine binds at every configuration and a closer approach buys nothing (the earlier ×26 and ×569 gains are withdrawn), while at the tagging optics a layer that follows the envelope is the whole difference between no tag and a 32–42% tag — the two are multiplicative. Figures 1–2 and Tables 1–2 re-derived (`nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py`, rectangular envelope per fragment azimuth); the sizing strip of §3 re-derived at the tagging envelope; §7 restated as the pricing of Report 1 §6.1; the energy ordering reversed with the reason given. The charge-identification material of §4, the obstacle table and the geometry correction of §6 are unchanged in substance and condensed.
2026-08-27	§7 replaced by a recovery study at the γ-matched energies: the lever table, the three levers dropped with reasons (the cold-beam argument runs backwards; dispersive tagging at IP6 is 30–60× short and the pipe cannot be instrumented past ≈ 30 m; the LHC 3σ is a pot window, the right precedent RHIC's β* = 22 m stores at 8σ). Earlier the same day: the charge argument restated as a sampled-Landau fake rate at matched efficiency (one bit costs 1.4×, a truncated mean 86× to one bit; the nanowire loses on fill factor), the α + d separation re-quoted from the

	repository's own density (median 10.9 / 30.1 / 73.4 mm), single-fragment loss identified as the dominant one-hit topology, EICROC's dynamic range promoted to the top of the unknowns.
2026-08-26	First version, revised the same day against an adversarial review that changed the sign of the charge-identification verdict: the incumbent AC-LGAD already digitises an 8-bit charge over 30 μm across four planes; r_s = 134 nm is a model parameter, the counting volume differs by 5000 between Argonne's papers, the ^{241}Am α tests √Q across β rather than z^2 at fixed velocity; the dead-edge argument retired at 0.005 mrad; the granularity argument redirected to ePIC's x-moving layer; the α + d topology added. Original content: the aperture scan and the chain at two apertures, the firing-threshold model, the obstacle table, the geometry correction, the question list.

All coherent acceptances use B = 50 GeV^{-2}; the near-beam envelope is a rectangle $10\sigma_h \times 10\sigma_v$ in angle at the Yellow Report divergences of Report 3 §4.1 (with the species step at 18 × 275) or at the tagging optics of Report 1 §6.1, applied to each fragment's azimuth; the measured silicon aperture is from `tools/fullsim` in the September-2024 `epic-main` (§6). Energy deposits are Bethe means in a 12 nm NbN film; the Landau most-probable value is not quoted because ξ/I ≈ 0.002 puts the film outside the ξ ≫ I regime. Hot-spot radii are anchored on Argonne's 134 nm and scaled as z. Built by `reports/build_report.py`; figures from the `evgen/scripts` named in the text; display mathematics typeset at build time.